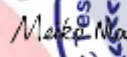


TEST REPORT EN 61960-3 Secondary cells and batteries containing alkaline or other non-acid electrolytes – Secondary lithium cells and batteries for portable applications –Part 3: Prismatic and cylindrical lithium secondary cells, and batteries made from them	
Report Number. :	CMC250303025
Date of issue..... :	2025-03-21
Total number of pages	13 Pages
Tested by (name, signature),	Jinsheng Li/Testing Engineer 
Reviewed by (name, signature) :	Meiko Ma /Project Manager 
Approved by (name, signature) :	Barry He/Technical Manager 
Name of Testing Laboratory preparing the Report	CMC Testing International (Shenzhen) Co., Ltd.
Address..... :	1-3/F., Building 7, He'er Hongben Industrial Zone, Dawangshan Community, Shajing Subdistrict, Baoan District, Shenzhen, Guangdong, China
Applicant's name	Xinxiang Hongli Supply Source Technology Co., Ltd.
Address..... :	West section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City , Henan Province
Manufacturer's name	Xinxiang Hongli Supply Source Technology Co., Ltd.
Address..... :	West section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang County, Xinxiang City , Henan Province
Test specification: Standard..... : EN 61960-3: 2017 (Partial test, only cl.7.2, cl.7.3.1, cl.7.3.2, cl.7.3.3 evaluated as client requested) Test procedure Type approved Non-standard test method N/A Test result..... : Pass	
Test item description..... : Cylindrical lithium-ion rechargeable cell Trade Mark..... : N/A Model/Type reference IMR18650-1200mAh Ratings 3.7V, 1200mAh, 4.44Wh	
General disclaimer: The test results presented in this report relate only to the object tested. This report shall not be reproduced, except in full, without the written approval of the CMC. The authenticity of this Test Report and its contents can be verified by contacting the CMC, responsible for this Test Report.	



List of Attachments (including a total number of pages in each attachment):

Attachment 1: Photo documentation (On pages 12).

Summary of testing:

Tests performed (name of test and test clause):

cl.7.2 Charging procedure for test purposes;
cl.7.3 Discharge performance;
cl.7.3.1 Discharge performance at 20°C (rated capacity);
cl.7.3.2 Discharge performance at -20°C
cl.7.3.3 High rate discharge performance

Partial test, only cl.7.2, cl.7.3.1, cl.7.3.2, cl.7.3.3 evaluated as client requested.

Testing location:

CMC Testing International (Shenzhen) Co., Ltd.

1-3/F., Building 7, He'er Hongben Industrial Zone,
Dawangshan Community, Shajing Subdistrict,
Baoan District, Shenzhen, Guangdong, China

Summary of compliance with National Differences (List of countries addressed):

N/A

☒ The product fulfils the requirements of EN 61960-3: 2017

Copy of marking plate:

Cylindrical lithium-ion rechargeable cell

IMR18650-1200mAh

3.7V, 1200mAh, 4.44Wh IMR19/66

Xinxiang Hongli Supply Source Technology Co., Ltd.

Date: YYYYMMDD Made in China

WARNING: Risk of Fire and Burns. Do Not Open, Crush, Heat Above 60°C/140°F or Incinerate. Do not short circuit. If bulges severely, discontinue use. Follow Manufacturer's Instructions.

Date Code: YYYYMMDD

YYYY=year, MM=month, DD=day

Test item particulars								
Electronic protection devices					Electrode plate			
End-of-discharge voltage					2.75V			
Manufacturer's declared internal a.c. resistance					<60 mΩ			
Manufacturer's declared internal d.c. resistance					<60 mΩ			
Possible test case verdicts:								
- test case does not apply to the test object					N/A			
- test object does meet the requirement					P (Pass)			
- test object does not meet the requirement					F (Fail)			
Testing								
Date of receipt of test item					2025-03-03			
Date (s) of performance of tests					2025-03-11 to 2025-03-17			
Test Environment Condition					Ambient temperature: 22.3°C~23.5°C			
Sample No.					SN250303025C001- SN250303025C013			
General remarks:								
The test results presented in this report relate only to the object tested. This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory. "(CXXX)" refers to sample number of cells, "X" is 0~9; "(See Enclosure)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report.								
Throughout this report a ☐ comma / ☒ point is used as the decimal separator.								
Name and address of factory (ies)					Same as applicant			
General product information:								
The cell consists of the positive electrode plate, negative electrode plate, separator, electrolyte, pouch. The positive and negative electrode plates are housed in the pouch in the state being separated by the separator.								
The main features of the cell in the battery pack are shown as below:								
Model (Cell)	Rated capacity	Nominal voltage	Nominal Charge Current	Nominal Discharge Current	Maximum Charge Current	Maximum Discharge Current	Maximum Charge Voltage	End Discharge Voltage
IMR186 50- 1200mA h	1200mAh	3.7V	240mA	240mA	600mA	600mA	4.2V	2.75V
The charging temperature range for this cell is: 0°C~45°C, discharging temperature is:-10~60°C								

EN 61960-3			
Clause	Requirement + Test	Result - Remark	Verdict
4	PARAMETER MEASUREMENT TOLERANCES		P
	Overall accuracy of controlled or measured values, relative to the specified or actual values, were within $\pm 1\%$ for voltage, current and capacity; $\pm 2\text{ }^{\circ}\text{C}$ for temperature; $\pm 0,1\%$ for time; $\pm 0,1\text{ mm}$ for dimensions		P
5	CELL DESIGNATION AND MARKING		P
5.1	Cell and battery designation		P
	Battery designated in the form of N1 A1 A2 A3 N2 / N3 / N4 – N5 in accordance with the conventions of this sub-clause	Complied.	N/A
	Cell designated in the form of A1 A2 A3 N2 / N3 / N4 in accordance with the conventions of this sub-clause	Complied.	P
5.2	Marking		P
	Cell or battery provided with clear and durable markings reflecting the following information	Complied.	P
	Secondary (rechargeable) Li or Li-ion..... :	Cylindrical lithium-ion rechargeable cell	P
	Battery or cell designation as in 5.1..... :	IMR19/66	P
	Polarity..... :	“+”, “-”	P
	Date of manufacture in real or code form..... :	See marking label.	P
	Name or identification of manufacturer or supplier..... ::	See marking label.	P
	Rated capacity..... :	See marking label.	P
	Nominal voltage..... :	See marking label.	P
	Caution for batteries which are considered swallowable	Not small battery.	N/A
	When battery surface is too small to accommodate all markings, information was given in specification sheet or in instruction manual or on the immediate package instead of on the battery		N/A
	Polarity need not be marked when the battery has been exclusively designed not to be connected wrongly		N/A
	Instead of designation as specified in 5.1, product code or name of battery was used when the battery is designed for use in particular end product		N/A
	By agreement between the cell manufacturer and battery and/or end product manufacturer, cells used in the assembly of a battery need not be marked		N/A

EN 61960-3			
Clause	Requirement + Test	Result - Remark	Verdict
	The battery designed not to be replaced by the end user also need not be marked		N/A
5.3	Providing the design and produce requirement of batteries		N/A
	Cell manufacturers provided equipment manufacturers who design and produce lithium-ion secondary batteries with the requirements specified in IEC 62133-2: 2017, Annex A		N/A
6	EXAMPLES OF CELLS		N/A
	Secondary lithium cell(s) suitable for standardization and used in assembling batteries complied with Table 2		N/A
7	ELECTRICAL TESTS		P
7.1	General		P
	The cell or battery samples used for tests are less than two months from the date of manufacture		P
	Charge and discharge currents for tests were based on the rated capacity (C5 Ah)		P
	Minimum values required for each electrical test were based on Table 5	Considered.	P
	Sample sizes and sequence of tests were in accordance with Figure 1	Considered.	P
7.2	Charging procedure for test purposes		P
	The cell or battery discharged at 20 °C ± 5 °C at a constant current of 0,2 It A down to a specified end-of-discharge voltage prior to charging	Discharge with 240mA.	P
	Cell or battery charged in an ambient temperature of 20 °C ± 5 °C using the method declared by the manufacturer, except as stated in this standard:		P
7.3	Discharge performance		P
7.3.1	Discharge performance at 20 °C (rated capacity)	Tested complied.	P
	Cell or battery charged in accordance with 7.2, and stored in an ambient of 20 °C ± 5 °C for 1 h to 4 h (h).....:		P
	Cell or battery discharged in an ambient of 20 °C ± 5 °C, at a constant current of 0,2 It A until its voltage was equal to specified end-of-discharge voltage (A) (V)		P
	The capacity (Ah) delivered above was not less than 100 % of the rated capacity declared by the manufacturer (Ah)	(see appended table 7.3.1)	P

EN 61960-3			
Clause	Requirement + Test	Result - Remark	Verdict
	The above two steps repeated up to four additional times, as necessary to meet this requirement (No. of tests)		P
7.3.2	Discharge performance at -20 °C	Tested complied.	P
	Cell or battery charged in accordance with 7.2, and stored in an ambient of -20 °C ± 2 °C for 16 h to 24 h (h).....		P
	Cell or battery discharged in an ambient of -20 °C ± 2 °C, at a constant current of 0,2 It A until its voltage was equal to specified end-of-discharge voltage (A) (V)		P
	Capacity (Ah) delivered during discharge performance test at -20 °C ± 2 °C (Ah).....	(see appended table 7.3.2)	P
7.3.3	High rate discharge performance at 20 °C	Tested complied.	P
	Cell or battery charged in accordance with 7.2, and stored in an ambient of 20 °C ± 5 °C for 1 h to 4 h (h).....		P
	Cell or battery discharged in an ambient of 20 °C ± 5 °C, at a constant current of 1,0 It A until its voltage was equal to specified end-of-discharge voltage (A) (V)		P
	Capacity (Ah) delivered during high rate discharge performance at 20 °C ± 5 °C (Ah)		P
7.4	Charge (capacity) retention and recovery		N
	Cell or battery charged in accordance with 7.2, and stored in an ambient of 20 °C ± 5 °C for 28 days		N
	Cell or battery discharged in an ambient of 20 °C ± 5 °C, at a constant current of 0,2 It A until its voltage was equal to specified end-of-discharge voltage (A) (V)		N
	Retained capacity (Ah) after 28 days of storage according to Table 5 (Ah)		N
	Cell or battery was then charged in accordance with 7.2 within 24 h following the above discharge test, and stored in an ambient of 20 °C ± 5 °C for 1 to 4 h (h)		N
	Cell or battery discharged in an ambient of 20 °C ± 5 °C, at a constant current of 0,2 It A until its voltage was equal to specified end-of-discharge voltage (A) (V)		N
	Recovery capacity according to Table 5 (Ah)		N
7.5	Charge (capacity) recovery after long term storage		N

EN 61960-3			
Clause	Requirement + Test	Result - Remark	Verdict
	Cell or battery charged in accordance with 7.2, and discharged in an ambient of $20\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$, at a constant current of 0,2 It A for 2,5 h		N
	Cell or battery stored in an ambient temperature of $40\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$, for 90 days		N
	Cell or battery charged in an ambient of $20\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$, using manufacturer's method		N
	Cell or battery stored in an ambient of $20\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ for 1 h to 4 h (h)		N
	Cell or battery discharged in an ambient of $20\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$, at a constant current of 0,2 It A until its voltage was equal to specified end-of-discharge voltage (A) (V)		N
	Capacity (Ah) delivered after long term storage according to Table 5 (Ah)		N
	The charging and storage steps, optionally, repeated up to four additional times as necessary to satisfy this requirement		N
7.6	Endurance in cycles		N
7.6.1	General		N
	Prior to charging, the cell or battery was discharged at $20\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ at a constant current of 0,2 It A, down to a specified end-of-discharge voltage		N
7.6.2	Endurance in cycles at a rate of 0,2 It A		N
	The total number of cycles obtained when the test is completed was not less than that specified for this characteristic in Table 5 (Ah) ...:		N
7.6.3	Endurance in cycles at a rate of 0,5 It A (accelerated test procedure)		N
	The following alternative procedures were carried out as an alternative to 7.6.2. The remaining capacity measured according to step 1 to step 3 of 7.3.1 when the test is completed was not less than that specified for this characteristic in Table 5 (Ah)		N
7.7	Battery internal resistance		N/A
7.7.1	General		N/A
	Internal resistance of a secondary lithium battery determined by either alternating current (AC) or direct current (DC) method		N/A
	AC method used first followed by DC method when there was a need to measure the internal resistance by both AC and DC methods on the same battery		N/A

EN 61960-3			
Clause	Requirement + Test	Result - Remark	Verdict
	Battery charged in accordance with 7.2 and stored in an ambient temperature of 20 °C ± 5 °C for 1 h to 4 h.....:		N/A
	Internal resistance measured in accordance with 7.7.2 or 7.7.3 in an ambient of 20 °C ± 5 °C		N/A
7.7.2	Measurement of the internal AC resistance		N/A
7.7.2.1	Measurement		N/A
	Alternating RMS voltage, U_a , measured while applying an alternating RMS current, I_a , at 1,0 kHz ± 0,1 kHz, to the battery, for a period of 1 s to 5 s		N/A
	Internal AC resistance, R_{ac} , calculated using the formula, $R_{ac} = \frac{U_a}{I_a}$		N/A
7.7.2.2	Acceptance criterion		N/A
	Internal AC resistance, R_{ac} of the battery was not greater than R_{ac} declared by the manufacturer		N/A
7.7.3	Measurement of the internal DC resistance		N/A
7.7.3.1	Measurement		N/A
	Battery discharged at a constant current of $I_1 = 0,2 I_t$ A, and at the end of a discharge period of 10 s ± 0,1 s, the discharge voltage U_1 under load measured and recorded		N/A
	Discharge current immediately increased to a value of $I_2 = 1,0 I_t$ A and the corresponding discharge voltage U_2 measured under load and recorded again at the end of a discharge period of 1 s ± 0,1 s		N/A
	Voltage measurements made at the terminals of the battery independent of the contacts used to carry current		N/A
	Internal DC resistance, R_{dc} , of the battery calculated using the formula, $R_{dc} = \frac{U_1 - U_2}{I_2 - I_1}$		N/A
7.7.3.2	Acceptance criterion		N/A
	Internal DC resistance of the battery was not great than R_{dc} declared by the manufacturer:		N/A
7.8	Electrostatic discharge (ESD)		N/A
7.8.1	General		N/A
	Ability of the battery to withstand electrostatic discharge evaluated on a battery containing electronic protection devices		N/A
7.8.2	Test procedure		N/A

EN 61960-3			
Clause	Requirement + Test	Result - Remark	Verdict
	Battery subjected to the tests in accordance with IEC 61000-4-2 for contact discharge at 4 kV and air discharge at 8 kV		N/A
7.8.3	Acceptance criterion		N/A
	The battery operated with all protection circuits operational		N/A

8	TEST PROTOCOL AND CONDITIONS FOR TYPE APPROVAL		P
8.1	Test protocol		P
	Sample size and protocol for conducting electrical tests in Clause 7 were in accordance with figure 1		P
8.2	Conditions for type approval		P
8.2.1	Dimensions		P
	Dimensions of cell or battery do not exceed the manufacturers' specified values		P
8.2.2	Electrical tests		N
8.2.2.1	Manufacturer's declared rated capacity of the cell or battery based on its performance under conditions of 7.3 to 7.8		N
8.2.2.2	All samples complied with the performance specified in 7.3 to 7.8		N
8.2.3	Conditional type approval		N
	Cell or battery considered conditionally type approved prior to the completion of the charge (capacity) recovery after storage test specified in 7.5 and the endurance in cycles test specified in 7.6.2 when		N
	a) 20 % of the required cycles of the endurance test were completed, and the capacity delivered during any discharge remains above 85 % of the rated capacity, and		N
	b) Requirements of all the other tests specified in Clause 7 were met		N

Table 7.3.1	Discharge performance at 20 °C (rated capacity) Batteries			P
Sample No.	Stored times (hour)	Discharge current (0.2ItA) (mA)	Discharge capacity (mAh)	Test-Result
SN250303025C001	1	240	1315.3	P
SN250303025C002	1	240	1312.2	P
SN250303025C003	1	240	1305.0	P
Supplementary information: benchmark (>100%, C5 Ah=2200mAh)				

Table 7.3.2	Discharge performance at -20 °C (Batteries)			P
Sample No.	Stored times (hour)	Discharge current (0.2ItA) (mA)	Discharge capacity (mAh)	--
SN250303025C004	16	240	728.8	P
SN250303025C005	16	240	739.8	P
SN250303025C006	16	240	746.1	P
SN250303025C007	16	240	747.4	
SN250303025C008	16	240	743.9	
Supplementary information: benchmark (>30%, C5 Ah=2200mAh)				

Table 7.3.3	TABLE: High rate discharge performance at 20 °C (Batteries)			P
Sample No.	Stored times (hour)	Discharge current (1.0 It A)(mA)	Discharge capacity (mAh)	Test-Result
SN250303025C009	1	1200	1296.4	P
SN250303025C010	1	1200	1303.0	P
SN250303025C011	1	1200	1302.0	P
SN250303025C012	1	1200	1302.7	P
SN250303025C013	1	1200	1303.7	P
Supplementary information: benchmark (>60%, C5 Ah=2200mAh)				

Table 7.7	TABLE: Battery internal resistance (Batteries)	N/A
Table 7.7.2	Measurement of the internal a.c. resistance	
Sample No.	$R_{ac}(m\Omega)$	Test-Result
--	--	--
--	--	--
--	--	--
Supplementary information: $R_{ac}=U_a/I_a$ The internal a.c. resistance of the battery shall be not greater than the value of R_{ac} , R_{ac} declared by the manufacturer.		

7.7.3	Measurement of the internal d.c. resistance					N/A
No.	$U_1(V)$	$U_2(V)$	$I_1(A)$	$I_2(A)$	$R_{dc}(m\Omega)$	Test-Result
--	--	--	--	--	--	--
--	--	--	--	--	--	--
--	--	--	--	--	--	--
Supplementary information: $R_{dc} = U_1 - U_2 / I_2 - I_1 (\Omega)$ The internal d.c. resistance of the battery shall be not greater than the value of R_{dc} , R_{dc} declared by the manufacturer.						

Attachment 1: Photo documentation



Picture 1



Picture 2

Important

1. The test report is invalid if it is not affixed the official seal of the laboratory to it.
2. Copies of the test report without the official seal of the laboratory are invalid.
3. It is forbidden to copy the test report partially without the written approval of the laboratory.
4. The test report is invalid without the signatures of Approver, Reviewer and Testing engineer.
5. The test report is invalid if it is blotted out.
6. Objections to the test report must be submitted to CMC within 15 days.
7. The test report is valid for the tested samples only.
8. As for the Verdict, "--" means "no need for judgement", "P" means "pass", "F" means "fail" and "N/A" means "not applicable".

Testing laboratory: CMC Testing International (Shenzhen) Co., Ltd.

Address: 1-3/F., Building 7, He'er Hongben Industrial Zone, Dawangshan Community, Shajing Subdistrict, Baoan District, Shenzhen, Guangdong, China

Tel: 400-1668-320

E-mail: info@cmczt-lab.com

<http://www.cmczt-lab.com>

-- End of Report --